Introduction to EXCEL-SOLVER

Solving linear programming problems with Excel

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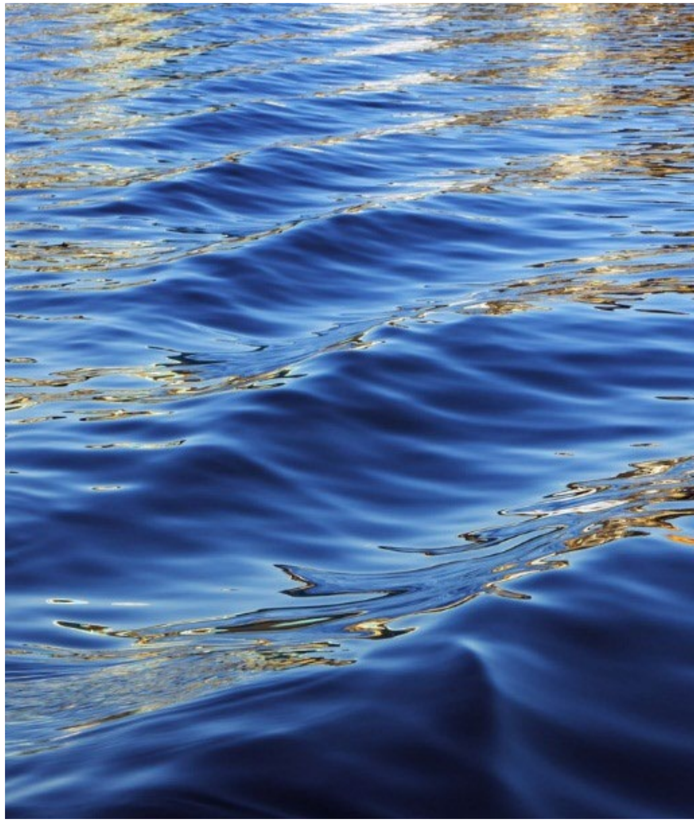

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Learning objectives

Learning objectives of this Chapter

- Learn how to use SOLVER for EXCEL

Methodology

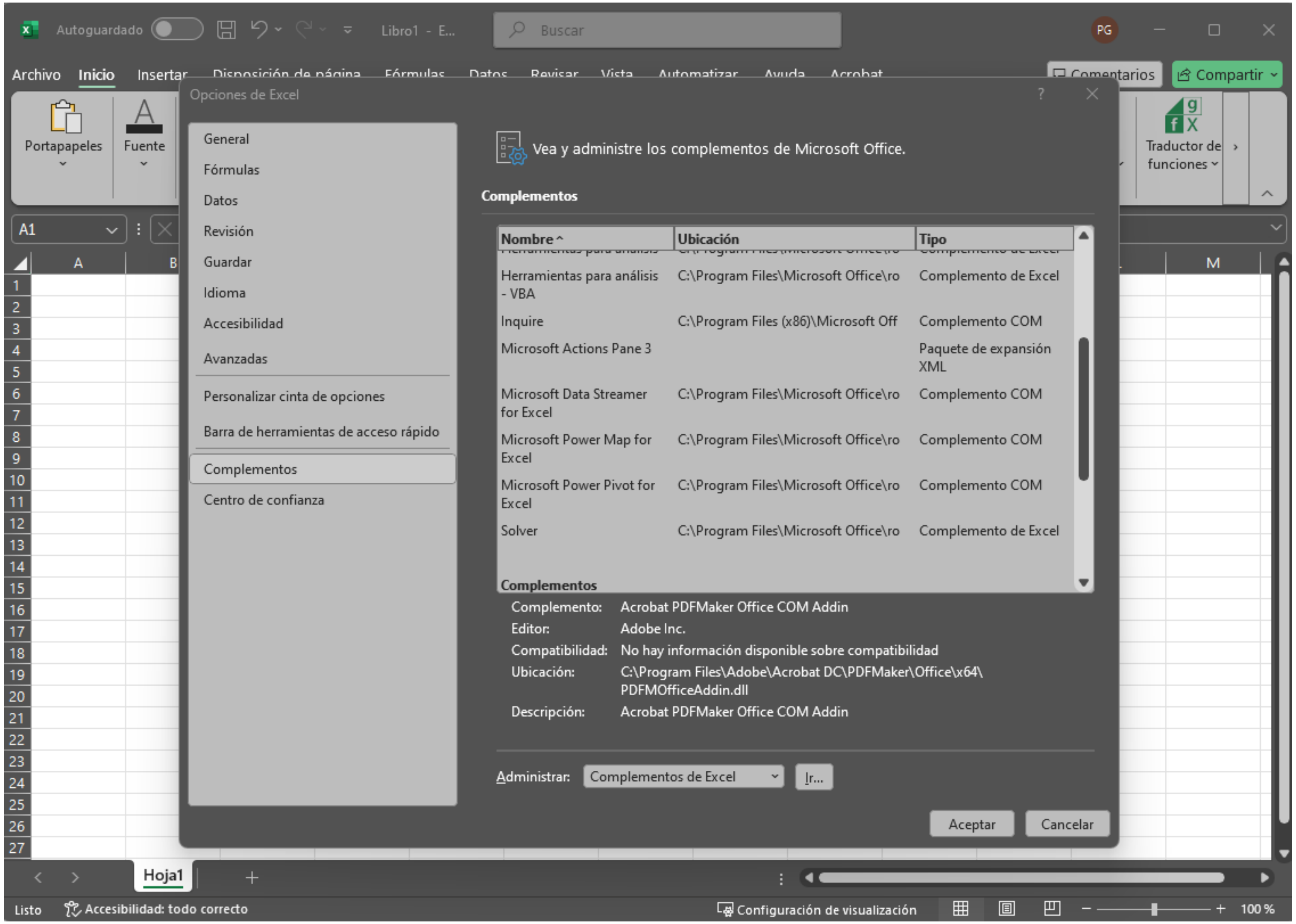
- We will use a practical example

What is SOLVER?

- Solver is an optimization tool in Excel that allows you to solve linear and nonlinear programming problems.
- It is used to find the optimal value (maximum or minimum) of an objective function, subject to constraints.
- It goes beyond Excel functions by implementing a true iterative search algorithm.

How to activate and use SOLVER in Excel?

1. Open Excel and go to the File tab.
2. Select Options.
3. Go to Complements.
4. In the Manage box, select Excel Add-ins and click Go.
5. Check the Solver box and click OK.



Limitations of Solver

- Small models → Maximum number of variables:
  - Up to 200 decision variables.
- Types of Models:
  - Mainly simple linear and nonlinear models.
- Resolution Time:
  - It depends on the complexity of the model and the hardware.
- Accuracy:
  - It can be affected by tolerance and accuracy settings.

Let's make an example of Linear Programming

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- **Objective**: Your company manufactures televisions, music players and BlueTooth speakers, using a common inventory of parts such as power supplies, speaker cones, etc.
  - Limited stock of components is available.
  - The most cost-effective set of products to build must be determined.
  - The following parts breakdown is available for each product:

			TV Set	Stereo	Speaker
			100	100	100
Part Name	Inventory	No. Used			
Chassis	450	200	1	1	0
Picture Tube	250	100	1	0	0
Speaker Cone	800	500	2	2	1
Power Supply	450	200	1	1	0
Electronics	600	400	2	1	1

  - The unit profit for the sale of each product is known:

		Profits:		
	By Product	75 €	50 €	35 €
	Total	16000		

Step 1: Create Excel tables for all known data

- Create an Excel table for the objective function coefficients and constraints.
- Enter the coefficients in the corresponding cells.

	A	B	C	D	E	F	G	H	I	J
1	**Example: Product Mix Problem**									
2	Your company manufactures TVs, stereos and speakers, using a common parts									
3	inventory of power supplies, speaker cones, etc. Parts are in limited supply and you									
4	must determine the most profitable mix of products to build. See our Tutorial Online									
5	for step-by-step instructions on formulating this linear programming model.									
6										
7										
8				TV Set	Stereo	Speaker				
9				100	100	100				
10	Part Name	Inventory	No. Used							
11	Chassis	450	200	1	1	0				
12	Picture Tube	250	100	1	0	0				
13	Speaker Con	800	500	2	2	1				
14	Power Suppl	450	200	1	1	0				
15	Electronics	600	400	2	1	1				
16				Profits:						
17			By Product	75 €	50 €	35 €				
18			Total	16000						
19										
20	To find the optimal solution click the Run button in the upper right of the Add-in.									
21	The optimal solution will appear within the worksheet. Press the Reports button to generate an Answer and Sensitivity report.									
22	The reports will be added as new worksheets in your workbook.									

Step 2: Configure the SOLVER tool on the basis of the data entered above.

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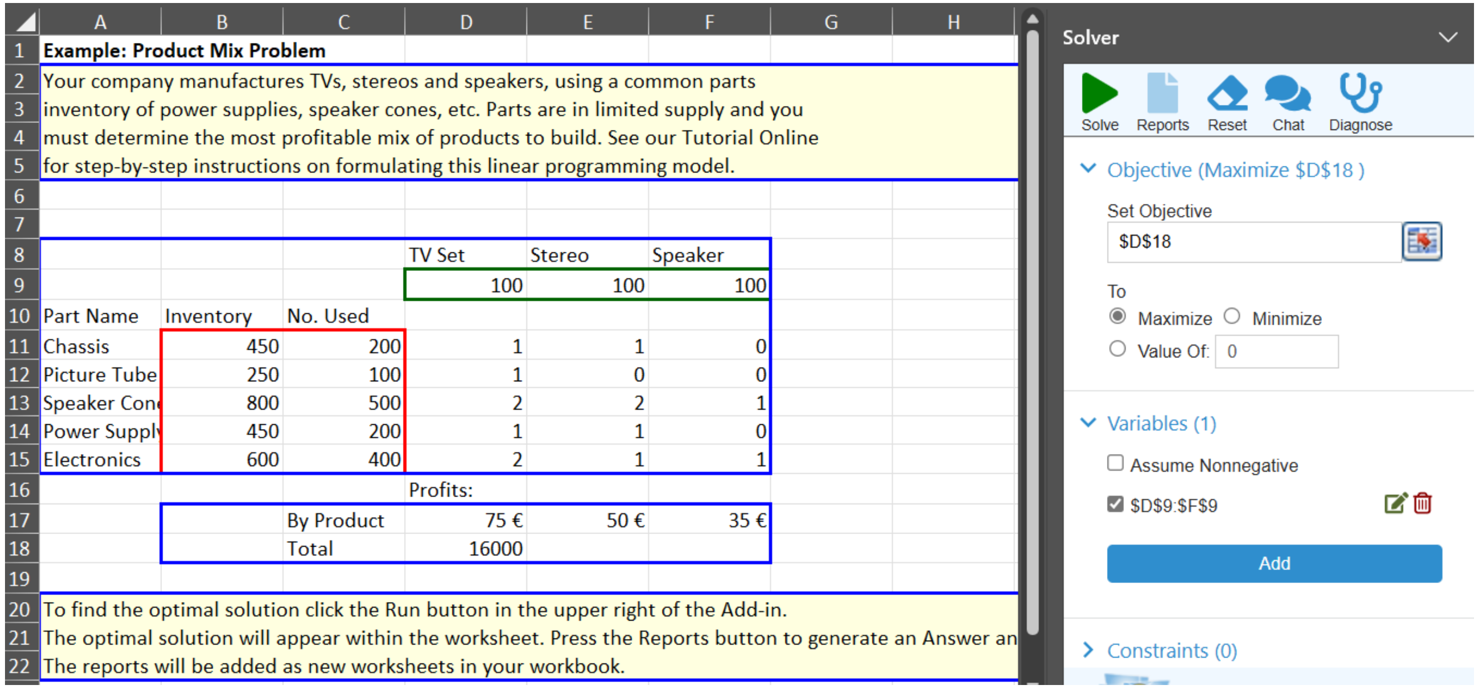
- Go to the Data tab and select Solver.
- 2. Define the target cell
- 3. Select Max to maximize the objective function.
- 4. Define the cells to be calculated
- 5. Add the restrictions using the "Add" option.



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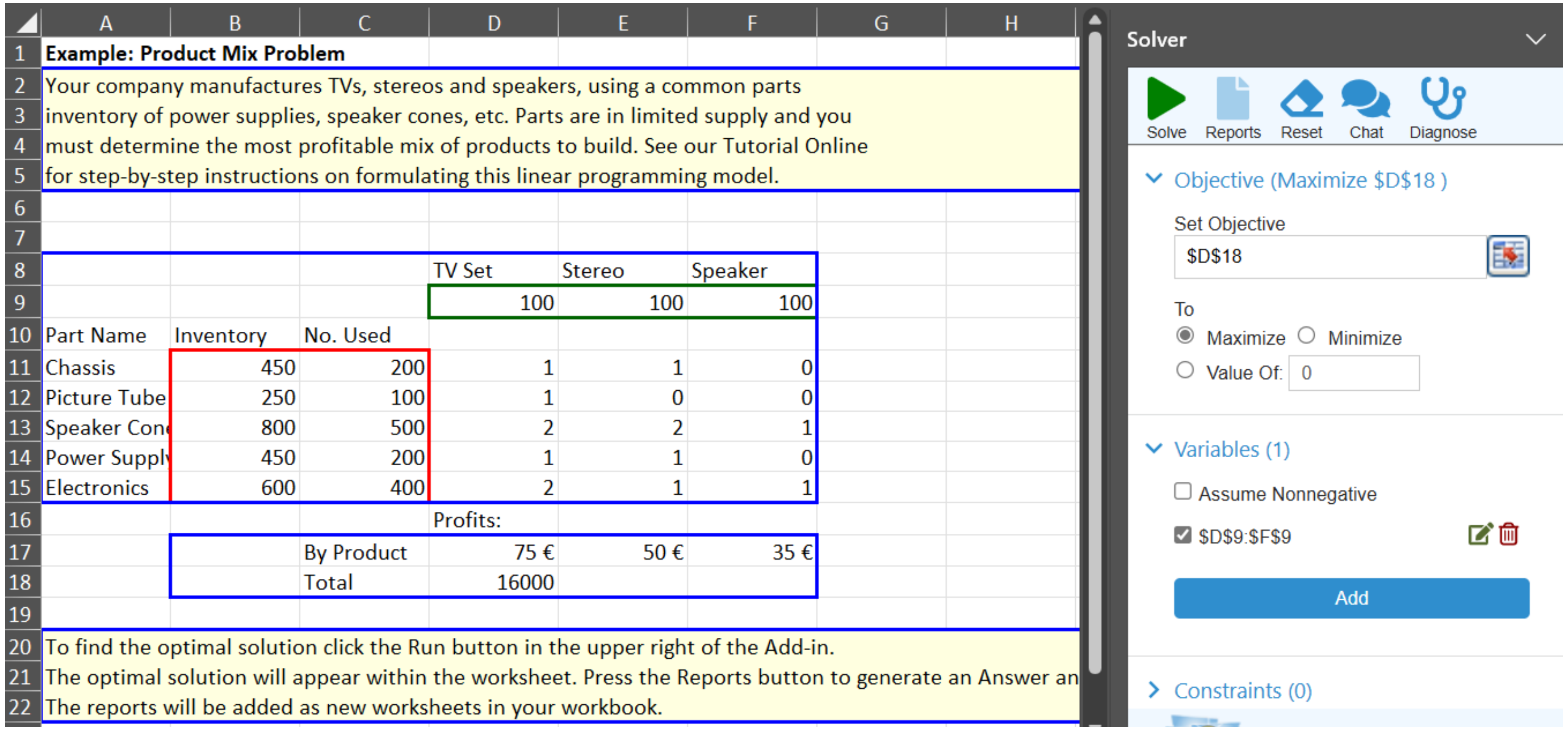
Step 3: run SOLVER and analyze the results

- Click on Solve.
- 2. Review the results provided by Solver.
- 3. Confirm that the solutions meet the constraints and the objective is optimal.



Conclusion

- Solver is a powerful tool for solving linear programming problems in Excel.
- It is important to correctly define the model and constraints.
- Recommended practice: Always check the results and validate with other methods if possible.